

Ameer

ogundeko Oluwatamilore



PAN-ATLANTIC
UNIVERSITY

DEPARTMENT OF ECONOMICS
SCHOOL OF MANAGEMENT AND SOCIAL SCIENCES
PAN-ATLANTIC UNIVERSITY, IBEJU-LEKKI, LAGOS
SECOND SEMESTER EXAMINATION, 2023/2024 ACADEMIC SESSION

ECO 104: Introductory Mathematics II (2 units)

Time Allowed: Two Hours

Instruction: Answer Question One and Attempt any other two questions of your choice.

Question One (25 marks) compulsory

1a. Matrix algebra is one of the rarely used mathematical tools in business. But it could be applied to several business areas such growth strategy amongst others. To aid strategic growth planning of firms, the ANSOFF matrix becomes handy. So, you are required to develop an ANSOFF matrix for any business of your choice showing the different growth strategies employed in each quadrant of its strategic plans. Also, identify four other possible practical areas of application of matrix algebra in business operations. [8 marks]

1b. A household consumes two electronic products- Samsung (X_1) and LG (X_2), the utility function from the consumption of these products is expressed as $U = 12X_1 + 18X_2 - 0.5X_1^2 - 0.5X_2^2$. If the income level of the household is \$60 and it costs \$2 and \$5 respectively to purchase a unit of Samsung and LG devices respectively. Required:

- Derive the household's marginal utility functions for Samsung and LG products respectively. [2 marks]
- With the aid of the Lagrangian multiplier, set up the constrained optimisation function and estimate the maximum utility of the household. [8 marks]
- Compute the value of λ and state its economic implications. [2 marks]

$$X_2 = 8.6897$$

$$X_1 = 8.2759$$

1c. How many terms in the GP 1, 14, 49, ... will be needed so that the sum of the first n terms is greater than 3355/4? [5 marks]

$$U = 1857$$

$$\lambda = 1.8621$$

Question Two [20 marks]

2a. Given that matrix $A = \begin{bmatrix} -9 & 8 & -0 \\ -7 & 4 & 5 \\ 1 & -2 & 3 \end{bmatrix}$, and $B = \begin{bmatrix} 1 & 4 & -5 \\ -4 & -15 & 20 \\ 5 & 18 & -24 \end{bmatrix}$

- Compute $|A|$ and $|B|$, state whether the matrix is a singular matrix or non-singular matrix with reason(s). [5 marks]
- Determine B^{-1} [5 marks]
- Compute $A \cdot B$ [3 marks]
- Proof that either $BB^{-1} = I$ [3 marks]

2b. A finance firm repays its mortgage of ₦662,500 by ₦1,000 in the first monthly instalment and then increases the payment by ₦500 in every instalment. How many instalments will it take the firm to pay off the mortgage? [4 marks]

Question Three [20 marks]

As policymakers worldwide reflect on the past three years since the global economy faced its most significant upheaval in the last 75 years, we find ourselves in a recovery phase marked by lingering challenges and varying growth trajectories across different regions. Given these circumstances, it is crucial to understand the roles of the product and money markets in the economy to strategically plan developmental programmes that will sustain fragile growth, especially in developing countries like Nigeria. Consider that the product and money markets interact to establish output level and interest rate that guarantee equilibrium and stability for sustained growth, the following mathematical models have been formulated for the economy: $C = 840 + 0.55Y$, $I = 286 - 35i$, $M_s = 450$, $M_d = 0.5Y$, $M_z = 104 - 50i$.

at (m)

Required:

- 3i. State the exact area of application of this mathematical case study in Economics. [2 marks]
- 3ii. Use the Cramer's rule to compute equilibrium income and interest rate of the economy and interpret your results with the aid of graphical illustration. [8 marks]
- 3iii. Estimate the speculative demand for money. [2 marks]
- 3iv. Calculate the value of savings. [2 marks]
- 3v. Determine the aggregate investment of the economy. [2 marks]
- 3vi. Comment on your results in 3iv and 3v above [2 marks]
- 3vii. Calculate the multiplier of the model and state its economic implications. [2 marks]

Question Four [20 marks]

Write short notes on the following mathematical concepts with practical examples, formulae, graphs, and areas of economic applications where possible.

- | | | |
|-------|--|-----------|
| i. | Fibonacci sequence and series. | [2 marks] |
| ii. | Convex function. | [2 marks] |
| iii. | Trigonometric functions. | [2 marks] |
| iv. | Inflection point. | [2 marks] |
| v. | Harmonic sequence and series. | [2 marks] |
| vi. | Geometric Sequence and series. | [2 marks] |
| vii. | Conformability conditions for matrix multiplication. | [2 marks] |
| viii. | Concave function. | [2 marks] |
| ix. | Constrained optimisation. | [2 marks] |
| x. | Cramer's rule. | [2 marks] |

Question Five [20 marks]

5a. A firm finds that the demand function for its output is $p = 420 - 0.7q^{0.5}$, what price should it charge to maximise sales revenue and what is the revenue maximizing quantity? [4 marks]

5b. Given the following set of linear simultaneous equations for the stock price of three firms in a competitive market. Use Cramer's to compute the values of each stock price.

$$2P_1 + 4P_2 - P_3 = 52$$

$$P_1 + 5P_2 + 3P_3 = 72$$

$$3P_1 - 7P_2 + 2P_3 = 10. \quad [7 \text{ marks}]$$

5c. Ebeano supermarket have the following cost function, $f(C) = 2 - 12q + 3q^2 + 2q^3$, estimate the value of total cost at the critical values, inflection point and test for concavity at the local extrema. [5 marks]

5d. An automobile is purchased for \$60,000. What will be its value after 10 years if it depreciates at the rate of 12% per annum? [4 marks]